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DRAINAGE IMPACT ANALYSIS

Halden Prairie, Revision 1

Prepared for Halden Prairie Development, LLC

Prepared by Ostrander Civil Group

August 21, 2026

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1.0 INTRODUCTION

The purpose of this report is to document the drainage design approach for the subject tract.

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EXECUTIVE SUMMARY

This analysis evaluates the effect of the proposed Halden Prairie development on downstream drainage and identifies the detention and conveyance needed to avoid increasing peak flows. Results are summarized in Sections 4 through 7.

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3.0 HYDROLOGIC METHODOLOGY

Runoff was computed for the 2-, 10- and 100-year events using the methods described herein.

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EXHIBIT

Drainage area map (graphic not reproduced in text).

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APPENDIX

Model input and output listings (not reproduced in text).

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4.2 FLOODPLAIN

The effective flood insurance rate map shows that approximately 3.1 acres along the southern boundary are within the effective floodplain. Any fill placed within this area will require equivalent mitigation storage and a floodplain development permit.

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5.1 OUTFALL

All onsite runoff will be collected and conveyed to the detention ponds before release. Due to its location within the flood control easement, the proposed outfall to Brannock Gully will require review and approval by the flood control district prior to approval of construction plans.

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6.1 PROPOSED CONDITIONS

Proposed land use was taken from the developer's planning documents. For this analysis, impervious cover of 58 percent was assumed based on the Rev B land plan, which is consistent with typical single-family development at this density.

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6.2 DETENTION REQUIREMENT

Applying the detention criteria for the receiving system to the full 62.4-acre tract, the development will require a minimum detention storage volume of 41.2 acre-feet. This volume accounts for the entire tract, including the commercial reserve.

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6.4 ALLOWABLE DISCHARGE

To avoid increasing peak flows in Brannock Gully, releases from the site will be limited to an allowable peak discharge of 18.7 cfs for the 100-year event. Release rates for the 2-year and 10-year events are provided in Table 6-3.

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2.0 EXISTING CONDITIONS

The tract is currently undeveloped pasture draining generally from north to south.

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7.1 DETENTION LAYOUT

Two detention ponds are proposed. Pond A, in the northeast portion of the site, provides 26.5 acre-feet of storage. Pond B is sized for 14.7 acre-feet with a bottom approximately 8 feet below natural ground. Side slopes and maintenance access are shown in Exhibit 7.

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8.1 FINISHED FLOOR ELEVATIONS

Based on the ponding elevations computed in Section 7, minimum finished floor elevations of 51.8 feet are recommended for all residential lots. Commercial pads will be evaluated separately when that reserve is designed.

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